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J-FB-22-00426

B. Tech. EXAMINATION, 2022

Semester VII (CBCS)

COMPUTER NETWORKS & DATA COMMUNICATION

EC-701

Time : 3 Hours *Maximum Marks :* 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt *Five* questions in all, selecting *one* question from each Sections A, B, C and D. Q. No. **9** is compulsory.

Section A

1. Explain TCP/IP Layered Model with a diagram. **10**

2. Write notes on the following :

(i) Layered Architecture of OSI Model **5**

(ii) LAN, WAN and MAN. **5**

Section B

3. Explain cyclic redundancy check. What are the advantages of cyclic coding ? **10**

4. Write notes on the following :

(i) Line and Block Coding **5**

(ii) Noiseless and Noisy Channel. **5**

Section C

5. Explain SONET/SDH network with a diagram. **10**

6. Write notes on the following :

(i) Fast Ethernet access method **5**

(ii) Hamming distance and interleaving. **5**

Section D

7. Explain IEEE 80.211 architecture. Compare Zigbee and Bluetooth. **10**

8. Write notes on the following :

(i) Wi-MAX Services **5**

(ii) Classful and Classless addressing. **5**

(Compulsory Question)

9. Explain the following : **4×5=20**

(i) Components of Data Communication

(ii) Latency and jitter

(iii) Block coding error

(iv) DWDM Network

(v) Layers in 802.16.